

MATHEMATICS

VOLUME AND SURFACE AREA MEMO

- 1.1 Each bump has a diameter of $1,8 \text{ m} \div 6 = 30 \text{ cm}$, ✓ so each radius = 15 cm . ✓

$$\begin{aligned} \text{For 6 bumps, metal needed} &= 6 \times \frac{1}{2} \left(\frac{4}{3} \pi (15)^3 \right) \checkmark \checkmark \\ &= 42\,411,50 \text{ cm}^3 \checkmark \end{aligned} \quad (5)$$

$$\begin{aligned} 1.2 \quad \text{Area to be painted} &= 3 \times \frac{1}{2} (4\pi(15)^2) \checkmark \checkmark \checkmark \\ &= 4\,241,15 \text{ cm}^2 \checkmark \end{aligned} \quad (4)$$

$$2.1 \quad H^2 = 1,5^2 + 1^2 \checkmark$$

$$\therefore H = 1,80 \text{ m} \checkmark$$

$$\begin{aligned} \text{Surface Area} &= \pi r^2 + \pi r H \\ &= \pi(1)^2 + \pi(1)(1,80) \checkmark \checkmark \\ &= 8,81 \text{ m}^2 \checkmark \end{aligned} \quad (5)$$

Also accept 8,79, obtained using rounded values.

$$2.2 \quad \text{Volume} = \text{Sphere} + 3 \text{ cubes}$$

$$\begin{aligned} &= \frac{4}{3} \pi r^3 + (3 \times l^3) \checkmark \\ &= \frac{4}{3} \pi (1,5)^3 + (3 \times 0,4^3) \checkmark \checkmark \\ &= 14,33 \text{ m}^3 \checkmark \checkmark \end{aligned} \quad (5)$$

$$3.1 \quad \text{Original cone} = \frac{1}{3}\pi(3)^2(4) \checkmark$$

$$= 37,69 \text{ cm}^3 \checkmark$$

$$\text{Smaller cone} = \frac{1}{3}\pi(2)^2(2) \checkmark$$

$$= 8,38 \text{ cm}^3 \checkmark$$

$$\therefore \text{Save approximately } 37,69 - 8,38 = 29,32 \text{ cm}^3 \checkmark$$

Also accept 29,31 if rounded values used.

(5)

$$3.2 \quad 380 = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) \checkmark$$

$$r = \sqrt[3]{\frac{570}{\pi}} \checkmark$$

$$= 5,66 \text{ cm} \checkmark$$

(3)

3.3 Would require a cone with a height of 4 cm and a radius of 5,66 cm.

$$\text{Cream required} = \frac{1}{3}\pi(5,66)^2(4) \checkmark$$

$$= 134 \text{ cm}^3 \checkmark$$

(2)

$$\begin{aligned}
 4.1 \quad \text{Area of rim} &= \pi(1,5)^2 - \pi(1)^2 \checkmark\checkmark \\
 &= 3,93 \text{ cm}^2 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Metal for rim} &= 3,93 \times 0,2 \\
 &= 0,79 \text{ cm}^3 \checkmark
 \end{aligned} \tag{4}$$

$$\begin{aligned}
 4.2 \quad \text{Volume of cone} &= \frac{1}{3}\pi(1)^2(3) \checkmark \\
 &= 3,14 \text{ cm}^3 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume of nail} &= \text{cone} + \text{rim} \\
 &= 3,14 + 0,79 \checkmark \\
 &= 3,93 \text{ cm}^3 \checkmark
 \end{aligned} \tag{4}$$

4.3 The volume of the rim is unchanged, so only the volume of the cone will change.

$$\begin{aligned}
 \text{New volume of cone} &= \frac{1}{3}\pi(1)^2(4) \checkmark \\
 &= 4,19 \text{ cm}^3 \checkmark
 \end{aligned}$$

$$\therefore \text{Will required an extra } 4,19 - 3,14 = 1,05 \text{ cm}^3 \text{ of metal. } \checkmark \tag{3}$$

$$\begin{aligned} 5.1 \quad \text{Volume} &= \pi(4)^2 \cdot (25) + \pi(4)^2 \cdot (30) + (3 \times 3 \times 3) + (3 \times 3 \times 6) + (3 \times 3 \times 9) \checkmark \checkmark \checkmark \checkmark \checkmark \\ &= 2\,926,60 \text{ mm}^3 \checkmark \end{aligned} \quad (6)$$

5.2 Amount to file off $(3 \times 3 \times 7) + (3 \times 3 \times 4) + (3 \times 3 \times 1) = 108 \text{ mm}^3 \checkmark \checkmark \checkmark$ (3)

5.3 Each key provides 108 mm^3 , so he needs $\frac{2926,60}{108} = 27,09 \checkmark$

So approximately 28 keys required. ✓

(27 is also acceptable, because $0,09 \text{ mm}^3$ is a negligible amount.) (2)

$$\begin{aligned} 6.1 \quad \text{Radius of shell} &= \frac{300 + 20 + 20}{2} \\ &= 170 \text{ cm } \checkmark \end{aligned}$$

Radius of hollow portion = 165 cm ✓

$$\begin{aligned} \text{Metal needed} &= \frac{1}{2} \left(\frac{4}{3} \pi (170)^3 \right) - \frac{1}{2} \left(\frac{4}{3} \pi (165)^3 \right) \checkmark \checkmark \\ &= 881\,478,54 \text{ cm}^3 \checkmark \end{aligned} \quad (5)$$

$$\begin{aligned} 6.2 \quad \text{Metal needed} &= 2 \times \pi \cdot (10)^2 (200) \checkmark \checkmark \\ &= 125\,663,71 \text{ cm}^3 \checkmark \end{aligned} \quad (3)$$

$$\begin{aligned} 6.3 \quad \text{Area to be painted} &= \left(\frac{1}{2} \times 4\pi r^2 \right) + (2 \times 2\pi rh) \\ &= \frac{1}{2} [4\pi (170)^2] + 2 \times 2\pi (10)(200) \quad \check{\check{\check{}}} \\ &= 206\,716,79 \text{ cm}^2 \quad \check \end{aligned} \tag{4}$$